

UNIT 2: E-COMMERCE BUSINESS MODEL (8HRS)

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Introduction to Business Model

- A business model is the method of doing business by which a company can sustain itself, that is, generate revenue. The business model spells out how a company makes money by specifying where it is positioned in the value chain.
- Some models are quite simple. A company produces goods or services and sells it to customers. If all goes well, the revenues from sales exceed the cost of operation and the company realizes profit. Other models can be more complex. Radio and television broadcasting is a good example. The broadcaster is part of a complex network of distributors, content creators, advertisers, and listeners or viewers. Who makes money and how much, It is not always clear at the outset. The bottom line depends on many competing factors.
- For our understanding, e-commerce can be defined as any form of business transaction in which the parties interact electronically.' A transaction in an electronic market represents a number of interactions between parties. For instance, it could involve several trading steps, such as marketing, ordering, payment, and support for delivery. An electronic market allows the participating sellers and buyers to exchange goods and services with the aid of information technology.

Introduction to Business Model

Business model solve the following questions:-

- What forces are driving the market place and industry in which business is competing?
- How business define it self?
- What are the competitive advantage that enable the business to prevail in a competitive marketplace?
- How does the business make money?

Electronic markets have three main functions such as:

- (i) matching buyers and sellers,
- (ii) facilitating commercial transactions, and
- (iii) providing legal infrastructure. Information technology permeates all the three functions and also helps to increase market efficiency and reduce transaction costs.

Introduction to Business Model

- The interaction between participants is supported by electronic trade processes that are basically search, valuation, payment and settlement, logistics, and authentication, as shown in Figure 2.1. The Internet and the World Wide Web allow companies to efficiently implement these key trading processes. For instance, many search services and brokers are available to help buyers find information, products, and merchants in electronic markets.

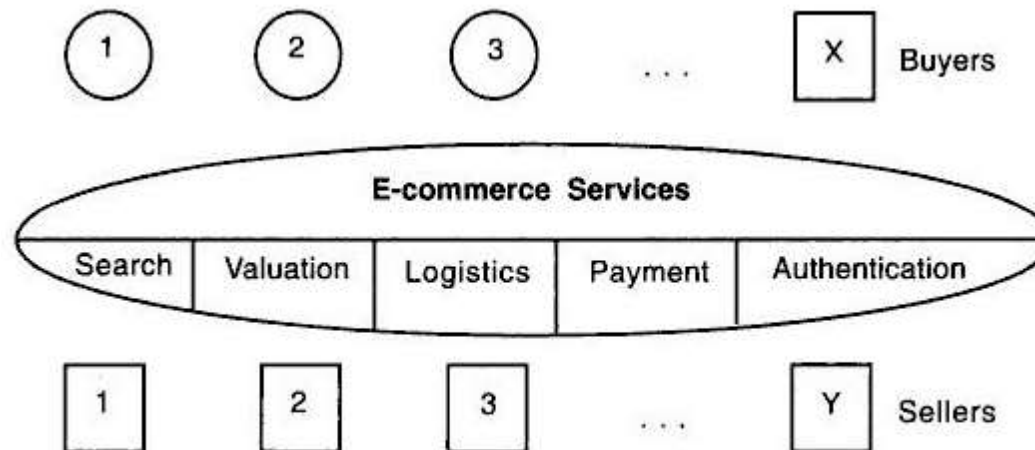


Fig. 2.1 Representation of an electronic market.

- There are various methods of doing business but we can classify the business model into two categories as:

a) E-Business models based on the relationship of Transaction Parties

- -Business to consumer (B2C)
- -Business to Business (B2B)
- -Consumer to Consumer (C2C)
- -Consumer to Business (C2B)

b) E-Business models based on the relationship of Transaction Types

- -Brokerage Model
- -Aggregator Model
- - Info-mediary Model
- -Value Chain Model
- -Community Model
- -Advertising Model

E-Business models based on the relationship of Transaction Parties

- Selling and buying process is directly and indirectly associate with various business parties and stakeholders to support business process. Basically, business is associated with the business we can classify on the basic of such parties. For example, an information B2C market differs in many respects from the automotive B2B market.
- The information B2C market represents companies that sell digital information goods, such as news, articles, music, books, or digital videos. In the information B2C market, the electronic infrastructure not only helps match customers and sellers, but also acts as the distribution channel, delivering products to customers.
- In the automotive B2B market, the products traded, such as parts and components of cars, have a high degree of specificity. The market infrastructure used is to be mainly based on Electronic Data Interchange (EDI) over expensive VAN services. EDI involves the exchange of standardized, structured information between organizations, permitting direct communication between computer systems. B2B is also a closed market in the sense that the number of participants involved in trading is limited and known a priori.

E-Business models based on the relationship of Transaction Parties

- Understanding the nature of the market's requirements is critical for creating the underlying e-business infrastructure. The relation between B2B and B2C models is clearly shown in Figure 2.3.

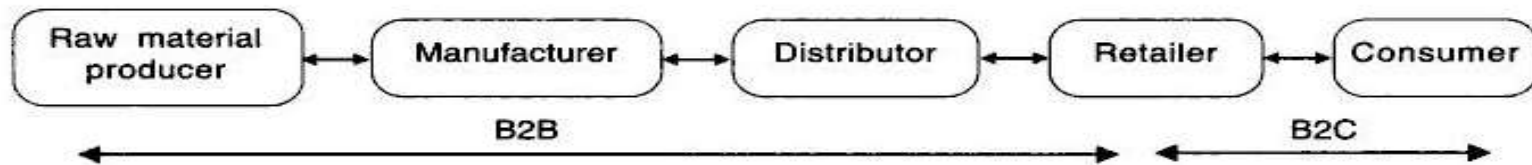


Fig. 2.3 Relation between B2B and B2C models.

B2B covers business transactions along the various interactions existing in the value chain from producers of raw materials to retailers and consumers including manufacturers and distributors.

On the contrary, B2C reflects only the interactions between a customer and a retailer. Basically, B2C transactions include the following steps:

- account acquisition.
- product discovery through search and browse,
- price negotiation,
- payment, and
- product delivery. In some cases, customer services may also exist.

E-Business models based on the relationship of Transaction Parties

TABLE 2.1
SUMMARY OF E-BUSINESS TRANSACTION MODELS

<i>Model</i>	<i>Description</i>	<i>Examples</i>
B2C	Sells products or services directly to consumers.	<i>amazon.com, autobytel.com, eDiets.com, Pets.com</i>
B2B	Sells products or services to other businesses or brings multiple buyers and sellers together in a central marketplace.	<i>MetalSite.com, VerticalNet.com, SHOP2gether.com</i>
B2G	Businesses selling to local, state, and federal agencies.	<i>iGov.com</i>
C2C	Consumers sell directly to other consumers.	<i>ebay.com, InfoRocket.com</i>
C2B	Consumers fix price on their own, which businesses accept or decline.	<i>Priceline.com</i>

E-Business models based on the relationship of Transaction Parties

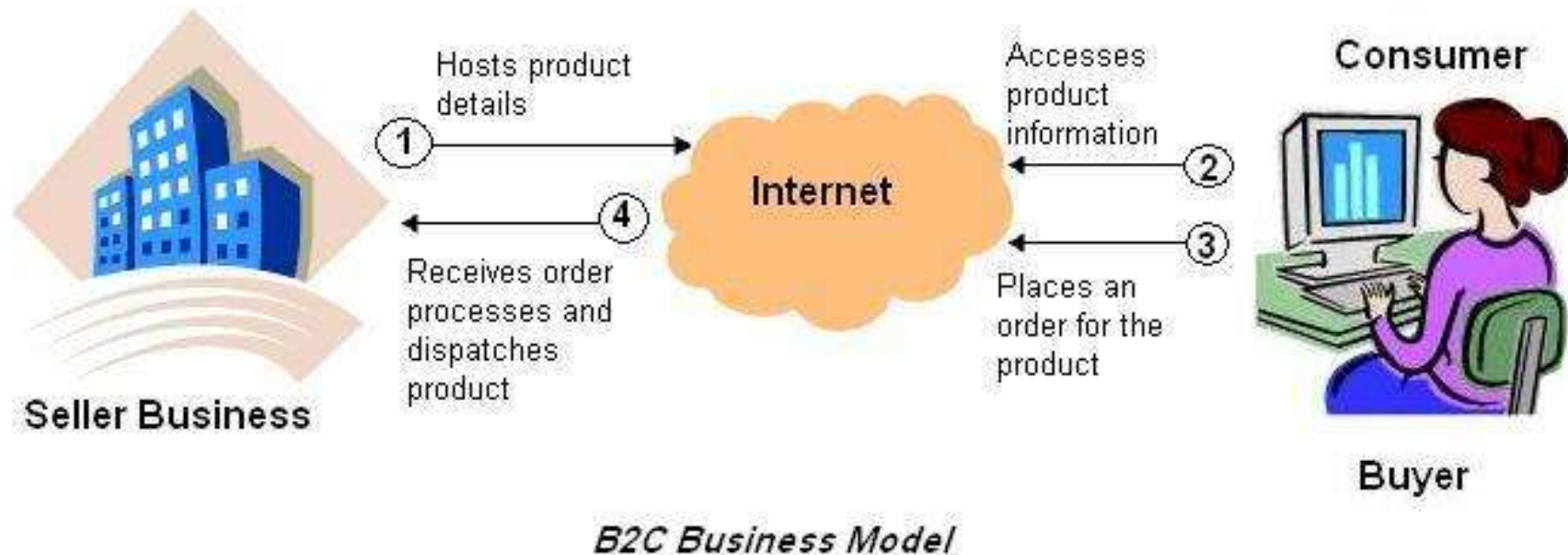
- E-commerce can be classified according to the transaction partners such as
 1. **business to-consumer (B2C),**
 2. **business-to-business (B2B),**
 3. **business-to-government (B2G),**
 4. **consumer to-consumer (C2C),**
 5. **consumer-to-business (C2B).**

E-Business models based on the relationship of Transaction Parties

1) Business-to-Consumer (B2C)

- The B2C model involves transactions between business organizations and consumers. It applies to any business organization that sells its products or services to consumers over the Internet. These sites display product information in an online catalog and store it in a database. The B2C model also includes services online banking, travel services, and health information and many more as shown in figure below.
-
- Consumers are increasingly going online to shop for and purchase products, arrange financing, arrange shipment or take delivery of digital products such as software, and get service after the sale. B2C e-business includes retail sales, often called e-retail (or e-tail), and other online purchases such as airline tickets, entertainment venue tickets, hotel rooms, and shares of stock.
- Some B2C e-businesses provide high-value content to consumers for a subscription fee. Examples of e-business following this subscription model include the Wall Street Journal (financial news and articles), Consumer Reports (product reviews and evaluations), and ediels.com (nutritional counseling).
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- B2C e-business models include virtual malls, which are websites that host many online merchants. Virtual malls typically charge setup, listing, or transaction fees to online merchants, and may include transaction handling services and marketing options. Examples of virtual malls include excite.com, choicemail, women.com, networkweb.com, amazon.com, Zshops.com, and yahoo.com.

E-Business models based on the relationship of Transaction Parties



E-Business models based on the relationship of Transaction Parties

Some of the reasons why one should opt for B2C are:

1. Inexpensive costs, big opportunities. Once on the Internet, opportunities are immense as companies can market their products to the whole world without much additional cost.
2. Globalization. Even being in a small company, the Web can make you appear to be a big player which simply means that the playing field has been leveled by e- business. The Internet is accessed by: millions of people around the world, and definitely, they are all potential customers.
3. Reduced operational costs. Selling through the Web means cutting down on paper costs, customer support costs, advertising costs, and order processing costs.
4. Customer convenience. Searchable content, shopping carts. promotions, and interactive and user-friendly interfaces facilitate customer convenience. Thus, generating more business. Customers can also see order status, delivery status, and get their receipts online.
5. Knowledge management. Through database systems and information management, you can find out who visited your site, and how to create, better value for customers.

Processes in B2C (How Does B2C Work?)

- B2C e-commerce is more than just an online store. It really is about managing the entire process, but just using technology as a tool for order processing and customer support. Figure 2.5 depicts the processes in B2C.

Processes in B2C (How Does B2C Work?)

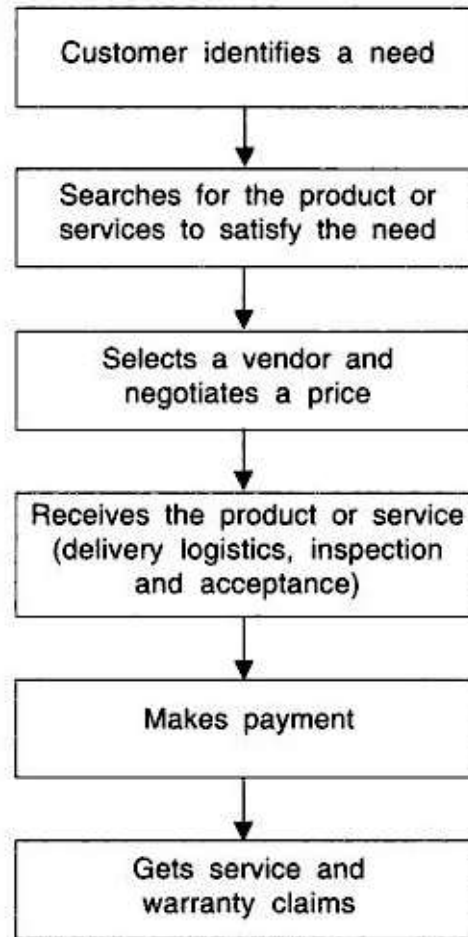


Fig. 2.5 Processes in B2C.

Processes in B2C (How Does B2C Work?)

The B2C process is now explained in greater details:

- **Visiting the virtual mall.** The customer visits the mall by browsing the online catalogue—a very organized manner of displaying products and their related information such as price, description, and availability. Finding the right product becomes easy by using a keyword search engine. Virtual malls may include a basic to an advanced search engine, product rating system, content management, customer support systems, bulletin boards, newsletters and other components which make shopping convenient for shoppers.
- **Customer registers.** The customer has to register to become part of the site's shopper registry. This allows the customer to avail of the shop's complete services. The customer becomes a part of the company's growing database and can use the same for knowledge management and data mining.

Processes in B2C (How Does B2C Work?)

- **Customer buys products.** Through a shopping cart system, order details, shipping charges, taxes, additional charges and price totals are presented in an organized manner. The customer can even change the quantity of a certain product. Virtual malls have a very comprehensive shopping system, complete with check-out forms.
- **Merchant processes the order.** The merchant then processes the order that is received from the previous stage and fills up the necessary forms.
- **Credit card is processed.** The credit card of the customer is authenticated through a payment gateway or a bank. Other payment methods can be used as well, such as debit cards, prepaid cards, or bank-to-bank transfers.
- **Operations management.** When the order is passed on to the logistics people, the traditional business operations will still be used. Things like inventory management, total quality management, warehousing, optimization and project management should still be incorporated even though it is an e-business. Getting the product to the customer is still the most important aspect of e-commerce.

Processes in B2C (How Does B2C Work?)

- **Shipment and delivery.** The product is then shipped to the customer. The customer can track the order/delivery as virtual malls have a delivery tracking module on the website which allows a customer to check the status of a particular order.
- **Customer receives.** The product is received by the customer, and is verified. The system should then tell the firm that the order has been fulfilled.
- **After-sales service.** After the sale has been made, the firm has to make sure that it maintains a good relationship with its customers. This is done through customer relationship management or CRM.

Processes in B2C (How Does B2C Work?)

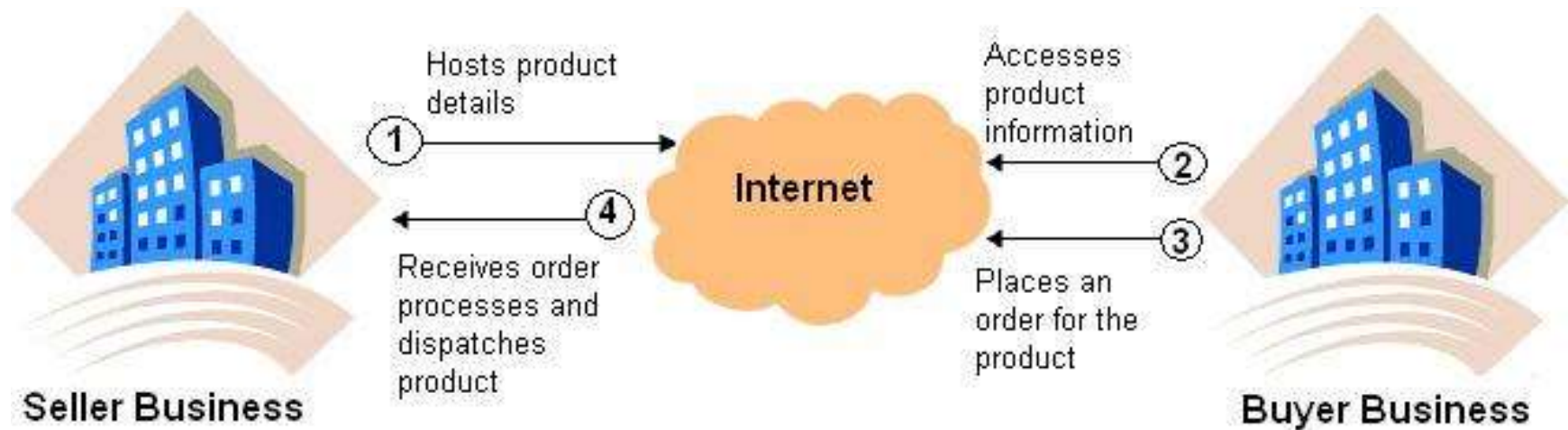
- The example of the www.amazon.com site also involves the B2C model in which the consumer searches for a book on their site and places an order, if required. This implies that a complete business solution might be an integration solution of more than one business model. For example, www.amazon.com includes the B2B model in which the publishers transact with Amazon and the B2C model in which an individual consumer transact with the business organization. The B2C model of e-commerce is more prone to the security threats because individual consumers provide their credit card and personal information on the site of a business organization. In addition, the consumer might doubt that his information is secured and used effectively by the business organization. This is the main reason why the B2C model is not very widely accepted. Therefore, it becomes very essential for the business organizations to provide robust security mechanisms that can guarantee a consumer for securing his/her information

E-Business models based on the relationship of Transaction Parties

2) Business to Business (B2B)

- The B2B model involves electronic transactions for ordering, purchasing, as well as other administrative tasks between business houses. It includes trading goods, such as business subscriptions, professional services, manufacturing, and wholesale dealings. Sometimes in the B2B model, business may exist between virtual companies, neither of which may have any physical existence. In such cases, business is conducted only through the Internet.
- Let us look at the example of www.amazon.com. As you know, www.amazon.com is an online bookstore that sells books from various publishers including Wrox, O'Reilly, Premier Press, and so on. In this case, the publishers have the option of either developing their own site or displaying their books on the Amazon site (www.amazon.com), or both. The publishers mainly choose to display their books on www.amazon.com as it gives them a larger audience. Now, to do this, the publishers need to transact with Amazon, involving business houses on both the ends, is the B2B model as shown in figure below.

E-Business models based on the relationship of Transaction Parties



B2B Business Model

E-Business models based on the relationship of Transaction Parties

- Thus, B2B is that model of e-commerce whereby a company conducts its trading and other commercial activity through the Internet and the customer is another business itself. This essentially means commercial activity between companies through the Internet as a medium.
- This is supposed to be a huge opportunity area on the Web. Companies have by and large computerized all the operations worldwide and now they need to go into the next stage by linking their customers and vendors. This is done by supply chain software, which is an integral part of your ERP application. Companies need to set up a backbone of B2B applications, which will support the customer requirements on the Web. Many B2B sites are company and industry specific, catering to a community of users, or are a combination of forward and backward integration. Companies have achieved huge savings in distribution-related costs due to their B2B applications.
- Major Advantages of B2B

E-Business models based on the relationship of Transaction Parties

1. **Direct interaction with customers.** This is the greatest advantage of e-business.
2. **Focussed sales promotion.** This information gives authentic data about the likes, dislikes and preferences of clients and thus helps the company bring out focussed sales promotion drives which are aimed at the right audience.
3. **Building customer loyalty.** It has been observed that online customers can be more loyal than other customers if they are made to feel special and their distinct identity is recognized and their concerns about privacy are respected. It has also been found that once the customers develop a binding relationship with a site and its product, they do not like to shift loyalties to another site or product.
4. **Scalability.** This means that the Web is open and offers round-the-clock access. This provides an access never known before, to the customer. This access is across locations and time zones. Thus a company is able to handle many more customers on a much wider geographical spread if it uses an e-business model. The company can set up a generic parent site for all locations and make regional domains to suit such requirements. Microsoft is using this model very successfully.
5. **Savings in distribution costs.** A company can make huge savings in distribution, logistical and after-sales support costs by using e-business models. Typical examples are of computer companies, airlines, and telecom companies.

Processes for Business-to-Business Transactions and Models

- B2B interactions involve much more complexity than B2C. For instance, typical B2B transactions include, among others, the following steps:
- review catalogues,
- identify specifications.
- define requirements,
- post request for proposals (REP).
- review vendor reputation.
- select vendor.
- fill out purchase orders (PO).
- send PO to vendor,
- prepare invoice,
- make payment,
- arrange shipment, and
- organize product inspection and reception.

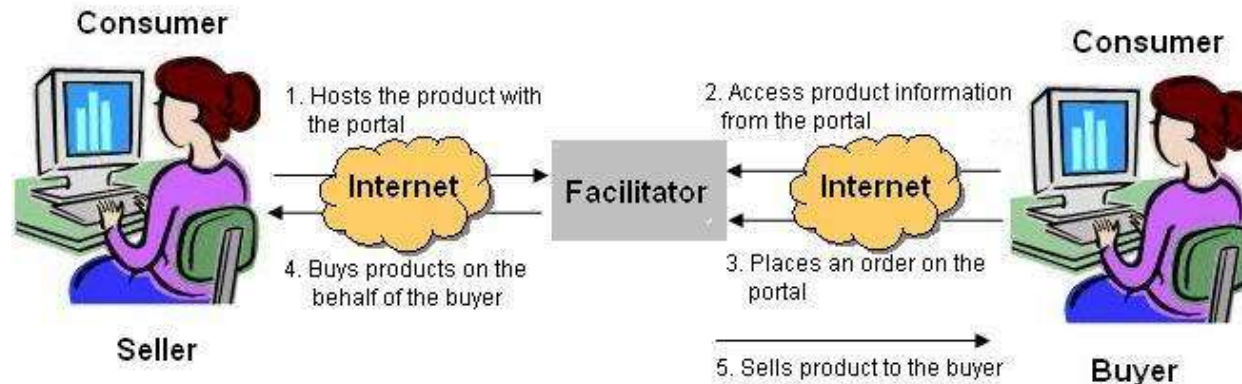
Processes for Business-to-Business Transactions and Models

- Due to the large number of transactions involved, business-to-business operations can be too risky if e-business sites cannot guarantee adequate quality of service in terms of performance, availability, and security.

E-Business models based on the relationship of Transaction Parties

3) Consumer to Consumer (C2C)

- The C2C model involves transaction between consumers. Here, a consumer sells directly to another consumer. eBay and www.bazee.com are common examples of online auction Web sites that provide a consumer to advertise and sell their products online to another consumer. However, it is essential that both the seller and the buyer must register with the auction site. While the seller needs to pay a fixed fee to the online auction house to sell their products, the buyer can bid without paying any fee. The site brings the buyer and seller together to conduct deals as shown in figure below.



C2C Business Model

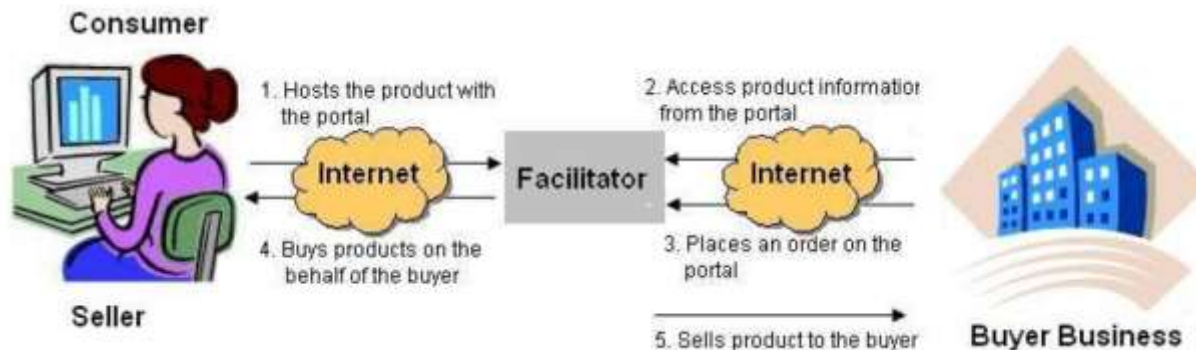
E-Business models based on the relationship of Transaction Parties

- Let us now look at the previous figure with respect to eBay. When a customer plans to sell his products to other customers on the Web site of eBay, he first needs to interact with an eBay site, which in this case acts as a facilitator of the overall transaction. Then, the seller can host his product on www.ebay.com, which in turn charges him for this. Any buyer can now browse the site of eBay to search for the product he interested in. If the buyer comes across such a product, he places an order for the same on the Web site of eBay. eBay now purchase the product from the seller and then, sells it to the buyer. In this way, though the transaction is between two customers, an organization acts as an interface between the two organizations.
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- There are also a number of new consumer-to-consumer expert information exchanges that are expected to generate \$6 billion in revenue by 2005. Some of these exchanges, such as AskMe.com and abuzz, are free, and some allow their experts to negotiate fees with clients. InfoRocket.com, one of the first question-and-answer marketplaces, is driven by a person-to-person auction format. The InfoRocket.com bidding system allows a person who submits a question to review the profiles of the "experts" who offer to answer the question. When the person asking the question accepts an "expert" offer, infoRocket.com bills the person's credit card, delivers the answer, and takes a 20 percent commission.

E-Business models based on the relationship of Transaction Parties

4) Consumer to Business (C2B)

- The C2B model involves a transaction that is conducted between a consumer and a business organization. It is similar to the B2C model, however, the difference is that in this case the consumer is the seller and the business organization is the buyer. In this kind of a transaction, the consumers decide the price of a particular product rather than the supplier. This category includes individuals who sell products and services to organizations. For example, www.monster.com is a Web site on which a consumer can post his bio-data for the services he can offer. Any business organization that is interested in deploying the services of the consumer can contact him and then employ him, if suitable as shown in figure.



E-Business models based on the relationship of Transaction Parties

- Let us look at another example of the C2B model. William Ward needs to buy an airline ticket for his journey from New York to New Jersey. William needs to travel immediately. Therefore, he searches a Web site for a ticket. The Web site offers bidding facility to people who want to buy tickets immediately. On the Web site, William quotes the highest price and gets the ticket.
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- In addition to the models discussed so far, five new models are being worked on that involves transactions between the government and other entities, such as consumer, business organizations, and other governments. All these transactions that involve government as one entity are called e-governance. The various models in the e-governance scenario are:
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E-Business models based on the relationship of Transaction Parties

- **Government-to-Government (G2G) model:** This model involves transactions between 2 governments. For example, if the American government wants to buy oil from the Arabian government, the transaction involved are categorized in the G2G model.
- **Government-to-Consumer (G2C) model:** In this model, the government transacts with an individual consumer. For example, a government can enforce laws pertaining to tax payments on individual consumers over the Internet by using the G2C model.
- **Consumer-to-Government (C2G) model:** In this model, an individual consumer interacts with the government. For example, a consumer can pay his income tax or house tax online. The transactions involved in this case are C2G transactions.
- **Government-to-Business (G2B) model:** This model involves transactions between a government and business organizations. For example, the government plans to build a fly over. For this, the government requests for tenders from various contractors. Government can do this over the Internet by using the G2B model.
- **Business-to-Government (B2G) model:** In this model, the business houses transact with the government over the Internet. For example, similar to an individual consumer, business houses can also pay their taxes on the Internet.

E-Business models based on the relationship of Transaction Types

- These transaction types take place in a variety of ways. Moreover, any given firm may combine one or two of these as part of its web business strategy.

1) Brokerage Model

- Brokers are market-makers: they bring buyers and sellers together and facilitate transactions. Brokers play a frequent role in business-to-business (B2B), business-to-consumer (B2C), or consumer-to-consumer (C2C) markets.
- The broker does not usually own the products.
- Usually, a broker charges a fee or commission for each transaction it enables. The formula for fees can vary depending on context.
- The **Brokerage Model** is an e-commerce business model where a **broker acts as a middleman** between buyers and sellers.
- Instead, it helps buyers and sellers find each other, complete transactions, and earns a **commission or service fee** for each successful transaction.

E-Business models based on the relationship of Transaction Types

- Brokerage models include:
 - **Marketplace Exchange** -- offers a full range of services covering the transaction process, from market assessment to negotiation and fulfillment. Some examples are [[Orbitz](#), [ChemConnect](#)].
 - Provides a complete marketplace where buyers and sellers can meet. Offers services such as product search, negotiation, ordering, and fulfillment.
 - **Example:**
A travel website where customers compare airline tickets and book flights.
- **Buy/Sell Fulfillment** -- takes customer orders to buy or sell a product or service, including terms like price and delivery. Accepts customer orders to buy or sell products. Handles pricing, ordering, and delivery arrangements.
- **Example:**
A customer orders a car online, and the broker connects the buyer with the dealer.
- Some examples are [[CarsDirect](#), [Respond.com](#)].
- **Auction Broker** -- conducts auctions for sellers (individuals or merchants). Broker charges the seller a listing fee and commission scaled with the value of the transaction. Auctions vary widely in terms of the offering and bidding rules. Some examples are [[eBay](#)]
- Conducts online auctions. Sellers list products for bidding.
- Buyers compete by placing bids. The highest bidder wins. The broker charges listing fees and commissions.

E-Business models based on the relationship of Transaction Types

- **Example Scenario:**

A seller lists an old laptop.

Many buyers bid.

The highest bidder purchases the laptop.

- **Transaction Broker** -- provides a third-party payment mechanism for buyers and sellers to settle a transaction. Some examples are [[PayPal](#), [Escrow.com](#)]

- Provides a secure payment system between buyers and sellers. Ensures safe money transfer. Charges a transaction fee.

- **Example:**

When purchasing from an online seller, PayPal securely processes the payment.

- **Search Agent** -- a software agent or "robot" used to search-out the price and availability for a good or service specified by the buyer, or to locate hard to find information.

- **Example Scenario:**

You search for an iPhone, and the search agent shows prices from different online stores.

- **Virtual Marketplace** -- or virtual mall, a hosting service for online merchants that charges setup, monthly listing, and/or transaction fees. It may also provide automated transaction and relationship marketing services. Some examples are [zShops and Merchant Services at [Amazon.com](#)].

- **Example:**

Small businesses sell products through Amazon instead of creating their own website.

E-Business models based on the relationship of Transaction Types

2) Aggregator Model

- The **Aggregator Model** is an e-commerce business model in which a company **collects (aggregates) information about products or services from multiple sellers** and displays it on a single website. The aggregator usually **does not manufacture or warehouse the products**. Instead, it helps customers compare products, prices, and features from different providers.
- Aggregator model includes:
- **Virtual Merchant** -- A **Virtual Merchant** is an online business that operates **only through the Internet**. It sells products or services without having a physical retail store. The method of selling may be listing price or auction. Some example includes [Amazon, eToys]
- **Example:**
A customer buys a laptop directly from Amazon's website without visiting a physical store.
- **Catalog Merchant** -- Customers browse an online catalog, select products, and place orders through the website.
- **Example:**
A clothing company displays all its products online with descriptions, prices, and images, allowing customers to order directly.

E-Business models based on the relationship of Transaction Types

- **Bit Vendor**
- A **Bit Vendor** sells **digital products and services** that can be delivered electronically over the Internet.
- Examples of digital products include:
 - E-books
 - Software
 - Music
 - Movies
 - Online courses
 - Digital games
- **Example:**
Purchasing an e-book or downloading software immediately after payment.
- **Subscription model** – the users have to pay for the access of the site. High value added content should be essential for subscription model. Some examples are [Wall street journal, Consumer Reports]
- **Examples:**
 - The Wall Street Journal (digital subscription)
 - Consumer Reports
- **Example:**
A user pays a monthly subscription to access premium news articles or research reports.

E-Business models based on the relationship of Transaction Types

3) Info-mediary Model

- The **Infomediary (Information Intermediary) Model** is an e-commerce business model where a company **collects, analyzes, and provides valuable information** about consumers, products, or markets.
- Unlike merchants, an infomediary usually **does not sell products directly**. Instead, it acts as an **information provider**, helping buyers make better purchasing decisions and helping sellers understand customer behavior.
- The collected information is valuable for:
 - Market research
 - Targeted advertising
 - Customer analysis
 - Business decision-making
- Info-mediary model includes:

E-Business models based on the relationship of Transaction Types

- **Advertising Networks** -- An **Advertising Network** connects advertisers with many websites.
- It displays advertisements (such as banner ads) across multiple websites.
- It tracks user behavior and collects data about visitors.
- Businesses use this data to measure advertising effectiveness and target the right audience.
- **Example:**
- DoubleClick (Google)
- **Example Scenario:**
You search for shoes online. Later, advertisements for shoes appear on other websites because the advertising network uses your browsing data.
- **Audience Measurement Services** -- These companies conduct **online market research** by measuring website traffic and audience behavior.
- They provide information such as:
- Number of visitors
- User demographics
- Popular websites
- User engagement
- Businesses use these reports to improve marketing strategies.
- **Example:**
- Nielsen//NetRatings

E-Business models based on the relationship of Transaction Types

- **Incentive Marketing** – In this model, customers receive **rewards** for shopping.
- Rewards may include:
 - Loyalty points
 - Coupons
 - Cashback
 - Gift vouchers
- The company also collects customer purchasing data, which can be analyzed and used for targeted advertising.
- **Example:**
 - CoolSavings
- **Example Scenario:**

A customer earns reward points after purchasing groceries online and later redeems those points for discounts.

E-Business models based on the relationship of Transaction Types

4) Community Model

- The **Community Model** is an e-commerce business model that is based on **user participation, loyalty, and interaction**. People join the platform because they share a **common interest**, such as education, technology, hobbies, or social networking.
- The success of this model depends on users spending their **time, effort, and emotions** in building and supporting the community.
- The platform earns revenue through:
 - Advertisements
 - Premium subscriptions
 - Selling related products or services
 - Voluntary donations
- **Open Source -- Open Source** software is developed collaboratively by programmers around the world. The source code is freely available. Anyone can use, modify, and improve the software. Developers voluntarily contribute to the project.
- **Examples:**
 - Red Hat Enterprise Linux
 - Linux
- **Example:**

Thousands of developers contribute to Linux by fixing bugs and adding new features.

E-Business models based on the relationship of Transaction Types

- **Open Content -- Open Content** refers to information that anyone can access, edit, and improve collaboratively.
- Contributors work voluntarily to create and maintain high-quality content.
- **Example:**
- Wikipedia
- **Example Scenario:**
People from around the world write and edit encyclopedia articles on Wikipedia.
- **Public Broadcasting** -- This model is commonly used by **non-profit organizations**.
- Instead of relying mainly on advertisements, the website is supported through **voluntary donations** from its community members.
- **Example:**
- The Classical Station (WCPE.org)
- **Example Scenario:**
Listeners donate money to keep an educational or public broadcasting website running.

E-Business models based on the relationship of Transaction Types

- **Social Networking Services** --These websites allow users to connect with other people who share similar interests.
- Users can:
 - Create profiles
 - Share photos and videos
 - Send messages
 - Join groups
 - Build personal or professional networks
- Revenue comes from:
 - Advertisements
 - Premium memberships
 - Business services
- **Examples:**
 - Facebook
 - Orkut (historical social network)

E-Business models based on the relationship of Transaction Types

5) Value Chain Model

- The **Value Chain Business Model** is an e-commerce model in which **products, services, or information move through different stages** from the **producer (manufacturer)** to the **final customer (end user)**.
- It not only manages the movement of products but also manages the **relationships** and **communication** between manufacturers, suppliers, distributors, retailers, and customers.
- A **value chain** is the sequence of activities involved in producing and delivering a product or service to customers.
- E-commerce supports the value chain through **two important business models**:
- **Demand Chain**
- **Supply Chain**

E-Business models based on the relationship of Transaction Types

- **Demand Chain**
- The **Demand Chain** focuses on the **customer side** of the business.
- Its goal is to understand customer needs and satisfy them by providing the right products and services.
- It includes activities such as:
 - Marketing
 - Customer relationship management (CRM)
 - Sales
 - Customer support
 - Online storefront
- **Example**
- A customer visits an online shopping website, orders a smartphone, pays online, and receives customer support after delivery.

E-Business models based on the relationship of Transaction Types

- **Supply Chain**
- The **Supply Chain** focuses on the **supplier side** of the business.
- It manages the movement of products from suppliers to manufacturers, distributors, retailers, and finally customers.
- It includes:
 - Purchasing raw materials
 - Manufacturing
 - Inventory management
 - Warehousing
 - Transportation
 - Delivery
- **Example**
- A smartphone manufacturer purchases chips, screens, and batteries from different suppliers before assembling and shipping the phones.

E-Business models based on the relationship of Transaction Types

Storefront and Hub

A. Storefront

- A **Storefront** is the online shop where customers can:
- Browse products
- Place orders
- Make payments
- Track deliveries
- **Purpose:** Directly sell products or services to customers.
- **Example:**
An online shopping website where customers purchase laptops or mobile phones.

B. Hub

- A **Hub** is a central platform used to communicate and coordinate with:
- Suppliers
- Manufacturers
- Distributors
- Business partners
- It helps businesses:
- Exchange information
- Process orders
- Manage inventory
- Coordinate deliveries
- **Purpose:** Manage relationships with partners and suppliers.

E-Business models based on the relationship of Transaction Types

6) Advertising Model

- The **Advertising Model** is an e-commerce business model in which a website provides **free or low-cost content and online services** to users while earning money by displaying **advertisements**.
- It is an extension of the traditional **TV, radio, and newspaper advertising model**, but on the Internet.
- In this model:
 - Users access the website for free.
 - Businesses pay the website to display advertisements.
 - The website earns revenue from those advertisements. The advertisements may appear as:
 - Banner ads
 - Pop-up ads
 - Video ads
 - Sponsored posts
 - Search ads
 - The advertising model is most successful when a website has:
 - A **large number of visitors**, or
 - A **specific target audience** (e.g., sports fans, students, or professionals).

E-Business models based on the relationship of Transaction Types

- **Examples**

- Some common websites that use the advertising model include:

- Google
- YouTube
- Facebook
- Online news websites
- Blog websites

- **Revenue Sources**

- A website may earn money through:
- Display/banner advertisements
- Pay-per-click (PPC) ads
- Sponsored content
- Video advertisements
- Affiliate advertising

E-Commerce Business/Revenue Model

- In business, revenue typically consists of the total amount of money received by the company for goods sold or services provided during a certain time period. Therefore, revenue models are a part of the business model.
- A revenue model is a framework to predict revenue performance and results over a period of time. Revenue models take into account the various ways your company generates revenue, also known as revenue streams.
- Many online companies generate revenues from multiple income streams such as advertising, subscription, marketing etc.
- There are five revenue models.
 1. Advertising Revenue Model
 2. Subscription Revenue Model
 3. Transaction Fee Revenue Model
 4. Sales Revenue Model
 5. Affiliate Revenue Model

1. Advertising Revenue Model

- The advertising revenue model is a business strategy where companies generate income by providing free or low-cost content, products, or services to users and monetizing that audience's attention by selling ad space to brands and marketers. An **advertising revenue model** is a **business** strategy where companies generate income by displaying advertisements to their audience.
- **Ad Inventory:** The total available space on a platform (websites, apps, or videos) where ads can be placed.
- **Ad Networks/Exchanges:** Intermediaries like Google AdSense or Meta Ads Manager that connect publishers with advertisers.

Common Pricing Models

- **CPC (Cost Per Click):** Advertisers pay only when a user interacts and clicks on their advertisement.
- **CPM (Cost Per Mille):** Advertisers pay a set fee for every 1,000 impressions (views) their ad receives.
- **CPA (Cost Per Action):** A high-risk, high-reward model where advertisers pay publishers only when an ad leads to a specific, measurable action (like a software download or a sale).
- **Advantages**
 - **High Scalability:** Ad revenue grows naturally alongside your user base and traffic.
 - **Wider Reach:** Offering a free tier or content funded by ads attracts a larger audience compared to paywalled alternatives.

2. Subscription Revenue Model

- The subscription revenue model is a framework where customers pay a recurring fee (weekly, monthly, or annually) to access a product or service.
- It shifts the focus from one-time transactions to long-term relationships, prioritizing continuous value delivery and customer retention.
- **1. Content or Access Subscriptions**
Customers pay a recurring fee to access digital platforms, services, or media content.
- WebToffee
- **Examples:** Video streaming (e.g., Netflix), music streaming (e.g., Spotify), or online news and software (SaaS).
- **2. Product or Box Subscriptions**
Customers receive physical goods delivered to their door on a recurring, scheduled basis.
- Stripe +1
- **Examples:** Meal kits, grooming products, or curated novelty items (e.g., Birchbox or HelloFresh).

Primary Benefits

- **High Customer Lifetime Value (LTV):** Cultivates repeat engagements and brand loyalty over time.
- **Better Financial Forecasting:** Gives businesses a reliable baseline to plan budgeting and scale operations.
- **Data-Driven Personalization:** Allows companies to study user habits and tailor their offerings continuously.

Subscription Revenue Model

- You must have heard of Netflix, Amazon Prime, YouTube Premium, etc who will let you enjoy their unlimited services.
- These eCommerce business models charge their users or rather subscribers based on a certain interval of time (daily, monthly or annual) to avail their services.
- The service offerings of these companies generally include music, videos, TV channels, magazines, special services, etc. which is offered to the subscribers for a price to watch/listen or get the latest edition

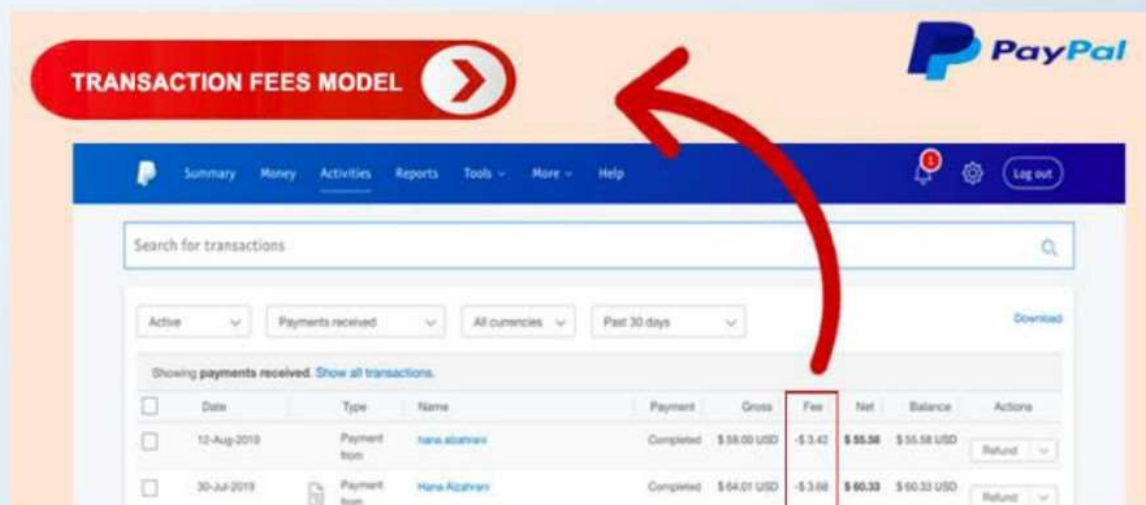


3. Transaction Fee Revenue Model

- A transaction fee revenue model is a pricing structure where a business charges a fee every time a transaction is processed, facilitated, or executed on its platform.
- Companies act as the middleman to connect buyers and sellers, taking a small cut—either as a flat rate or a percentage—for each successful sale.
- This model is prevalent in online marketplaces, payment processing services, and various financial platforms.
- **Percentage-Based:** The company takes a defined percentage of the total transaction value (e.g., 2% to 4%).
- **Flat Rate:** A fixed, set fee is charged per transaction, regardless of the order size.
- **Hybrid:** Combines both, such as charging a fixed rate plus a percentage (e.g., \$0.30 + 2.9%).
- **Payment Processing:** Gateways and processors like Esewa or PayPal charge businesses for handling payment routing and security.

Transaction Fee Revenue Model

- The eCommerce business following the transaction fee revenue model charges a fee to a seller for every transaction made through them.
- They are the payment companies that provides the payment gateway service to other eCommerce business platforms.
- Generally, the profit is derived through enabling or executing transactions



TRANSACTION FEES MODEL

PayPal

Summary Money Activities Reports Tools More Help

Search for transactions

Active Payments received All currencies Past 30 days Download

Showing payments received. Show all transactions.

	Date	Type	Name	Payment	Gross	Fee	Net	Balance	Actions
☐	12-Aug-2019	Payment from	Hana alsharrah	Completed	\$ 58.00 USD	-\$ 3.42	\$ 55.58	\$ 55.58 USD	Refund
☐	30-Jul-2019	Payment from	Hana Alsharrah	Completed	\$ 64.01 USD	-\$ 3.68	\$ 60.33	\$ 60.33 USD	Refund

4. Sales Fee Revenue Model

- The **Sales Fee Revenue Model** is an e-commerce business model in which a company **earns revenue by charging a fee or commission each time a product or service is sold through its platform**. The platform connects buyers and sellers and takes a percentage or fixed fee from every successful transaction.
- **How it Works**
 - Sellers list their products or services on the platform.
 - Customers purchase the products.
 - The platform processes the transaction.
 - The platform deducts a sales fee or commission and transfers the remaining amount to the seller.
- **Revenue Source**
 - Commission on each sale (e.g., 5–20% of the selling price)
 - Fixed transaction fee
 - Payment processing fee

4. Sales Fee Revenue Model

- **Examples**

- eBay – Charges sellers a fee for each completed sale.
- Amazon – Collects referral fees from third-party sellers.
- Etsy – Earns a transaction fee on every sale made by sellers.
- Airbnb – Charges service fees to hosts and guests for each booking.

- **Advantages**

- Generates steady revenue as sales increase.
- Sellers pay only when they successfully make a sale.
- Encourages the platform to attract more buyers and sellers.
- Easy to scale with increasing transaction volume.

Simple Example

- Suppose a seller lists a laptop for **\$1,000** on an online marketplace. If the platform charges a **10% sales fee**:
- Selling Price = **\$1,000**
- Sales Fee (10%) = **\$100**
- Seller Receives = **\$900**
- Platform Revenue = **\$100**

Sales Revenue Model

- This is the most commonly followed eCommerce business model where wholesalers and retailers sell their product over the internet intending to reach out to a larger target audience.
- often competitive in comparison to the actual store price.
- The business following the online sales model often comes with marketplaces as common entry points that allow them to deal with various product vendors allowing them to grow the marketplace and therefore earn more



5. Affiliate Model revenue model

- The **Affiliate Revenue Model** is a business model in which a website or online platform **earns a commission by promoting another company's products or services**. When a visitor clicks an affiliate link and completes a purchase or another required action (such as signing up), the affiliate receives a commission.

How it Works

- A company creates an affiliate program.
- Affiliates (bloggers, influencers, or websites) join the program.
- The affiliate shares special referral (affiliate) links.
- Customers click the link and buy a product or complete a required action.
- The company pays the affiliate a commission.

Examples

- Amazon – Affiliates earn commissions by promoting products through the Amazon Associates program.
- Booking.com – Travel websites earn commissions by referring hotel bookings.
- Shopify – Pays affiliates for referring new merchants.
- Technology review blogs and YouTube channels often earn commissions by recommending products and including affiliate links.

5. Affiliate Model revenue model

- **Advantages**

- No need to create or manage products.
- Low startup cost.
- Can generate passive income.
- Businesses increase sales through affiliate marketing.

- **Disadvantages**

- Income depends on website traffic and conversions.
- Commission rates may be low.
- Affiliates have little control over the merchant's pricing or products.
- Affiliate program rules and commission rates may change.

- **Simple Example**

- A blogger writes a review of a smartphone and includes an Amazon affiliate link.
- Smartphone Price = **\$500**
- Affiliate Commission = **5%**
- Customer purchases through the affiliate link.
- Affiliate Earnings = **\$25**
- Amazon receives the remaining revenue from the sale.

Affiliate Revenue Model

- is a revenue model which deals with a business that follows the principle of commission.
- Merchants and vendors partner up with well-known eCommerce platforms to advertise and sell their product giving them a percentage of the profit as a commission.
- An affiliate marketing is a well-known way of inviting as well as driving quality leads into their business.



Electronic Data Interchange (EDI)

- Electronic Data Interchange (EDI) is a technology that allows businesses to exchange standardized electronic documents directly between computer systems.
- By moving from a paper-based exchange of business document to one that is electronic, businesses enjoy major benefits such as reduced cost, increased processing speed, reduced errors and improved relationships with business partners.
- In EDI transactions, information moves directly from a computer application in one organization to a computer application in another.
- EDI standards define the location and order of information in a document format.
- Sharing data is now faster and easier with this automated capability, compared to using paper documents or other traditional methods that can take hours, days or even weeks.
- Today, industries use EDI integration to share a range of document types, from purchase orders to invoices to requests for quotations to loan applications and more.
- In most instances, these organizations are trading partners that exchange goods and services frequently as part of their supply chains and business-to-business (B2B) networks.
- E-payment systems eliminate the need for cash and paper checks.

Benefits of EDI

- The advantages of electronic transmission over paper, fax, and e-mail processes were numerous and compelling:
- **Greater efficiency** – replacing slow and cumbersome manual processes with faster electronic workflows
- **Fewer errors** – minimizing the risk of errors due to data entry mistakes, illegible handwriting, or miscommunication
- **Improved data accuracy** – with predefined standards, verification processes, and data consistency
- **Faster transaction processing and integration** – resulting in greater responsiveness to customer demands and faster decisions
- **Cost savings** – reducing costs related to document production, mailing, and storage as well as significant staffing costs related to time-consuming manual processes
- **Compliance** – helping organizations comply with industry regulations by adhering to standards
- **Environmental benefits** – radically decreasing the use of paper and preserving trees

•

- **Reduced costs** by eliminating paper, printing, and postage expenses.
- **Faster processing** of business transactions.
- **Fewer errors** because manual data entry is minimized.
- **Improved accuracy** through standardized document formats.
- **Better relationships** with business partners due to quicker and more reliable communication.
- **Increased efficiency** in supply chain and procurement processes.

Simple Example

- A retail store needs to order products from a supplier.
- **Without EDI:** The retailer prepares a paper purchase order, mails or emails it, and the supplier manually enters the information into its system.
- **With EDI:** The retailer's inventory system automatically generates a purchase order and sends it electronically to the supplier's computer system. The supplier's system processes the order immediately without manual intervention.
- This automation saves time, reduces errors, and speeds up the entire ordering process.

How EDI works?

To enable machines to exchange information, data must be structured in a format that both the sending machine and the receiving one can understand. Standardized formats and communication p

There are several steps in the EDI process:

- **Document creation.** The sender creates a document to be shared – a purchase order, invoice, shipping notice, or another document.
- **Translation.** Before it can be transmitted electronically, the document must be translated into a standardized EDI format. This translation is typically handled by EDI software.
- **Transmission.** The translated document can now be transmitted to the trading partner. Various communication methods can be used for transmission, including EDI file transfer protocol (FTP) and web-based methods.
- **Data receipt.** At this step, the receiving organization's EDI system receives the document, and it's the recipient's responsibility to check for completeness as well as compliance with agreed-upon EDI standards.
- **Translation – receiver side.** The EDI document is now ready to be translated to a format the recipient can read, so it can be processed by the organization's internal systems.
- **Processing.** Now, the data from the translated document can be automatically processed, thanks to integration into the recipient's ERP and other systems. This integration reduces the need for manual data entry and improves accuracy.
- **Confirmation.** In many cases, EDI messages are sent back to the sender to advise that the document was received and processed. EDI systems also often include archiving features to store past transactions, which is important for record-keeping and compliance purposes.

How does EDI work ?

Sending
Company

Receiving
Company

Data in
Internal
Format

EDI
translator

EDI
Data

Communications
Software

1. Prepare Document to be sent.

2. Translate the Document into EDI format.

3. Connect and transmit your document to your business partner.

Data in
Internal
Format

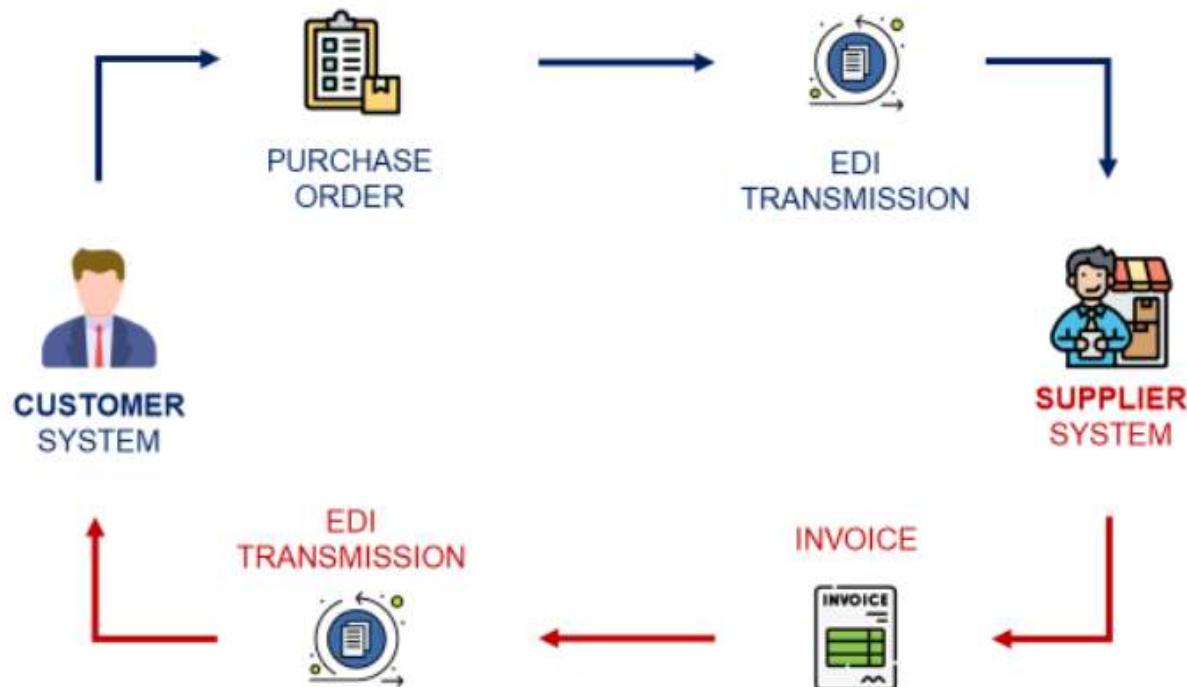
EDI
Translator

EDI
Data

Communications
Software

Direct, Point to Point
Connection

- As the supply chain becomes more digital, effective data exchange has become a must-have for any major company.
- Electronic Data Interchange (EDI) was created to facilitate efficient, reliable, and secure data exchange.
- After several decades, it has deeply imposed itself as a must-have for any large modern business.
- It facilitates the automatic transmission of business documents in a standardized format.



- In the example above, a purchase order is translated into an EDI message for transmission.
- Order is created by the **purchasing team** and received by the **supplier**
- Order information includes customer, supplier, delivery address and date, invoice address and detailed information about ordered items
- Invoicing, delivery and company information are mapped **using IDs** (Company ID, Location ID, ...)

PURCHASE ORDER				
Company:		Delivery Address:		
HIDI-EYE RETAIL CO.		Regional DC Paris		
Purchasing Department		21 rue de		
71 boulevard Maïsherbes		92100, Paris		
75017, Paris				
Supplier:		Invoice Address:		
Rizzi Tasse Manufacturing		HIDI-EYE RETAIL CO.		
21 rue Jean Jaures		Accounting Department		
95100, Argenteuil (France)		71 boulevard Maïsherbes		
		75017, Paris		
Order Creation Date:		Requested Delivery Date:		
Paris, January, 11 2018		January 11, 2018 10:00 am		
ID	EAN	Article	Unit	Quantity
1	9012345123451	Product 1	Pal	14
2	9012345757588	Product 2	Pal	23
3	9012345453749	Product 3	Pal	12
Order Number: 11424				

EDIFACT / EANCOM ORDER
UNB+UNOA:3+9099999000055:14+9012345000042:14+160111:1014+458893++ORDERS'
UNH+3654+ORDERS:D:96A:UN:EAN008'
BGM+220+11424+9
DTM+137:20180111:102,
DTM+2:20180112000:203'
NAD+BY+9099999000055::9'
NAD+SU+9012345000042::9'
NAD+DP+90999990000178::9'
NAD+IV+9099999000055::9'
LIN+1++9012345123451:EN'
QTY+21:14'
LIN+2++9012345757588:EN'
QTY+21:23'
LIN+3++9012345463749:EN'
QTY+21:12'
UNS+S'
UNT+16+3654'
UNZ+1+458893'

Elevate

DashboardOrdersEDI TransactionsEDI InterchangesReportsConfiguration

Purchase Orders

Shipments

Invoices

Adjustments

Purchase Order 12345

New

Close

Save

Accept PO

Shipments

Invoices

Send to Acumatica

Delete

PO #: 12345

Date: 2025-01-28

Received: 2025-06-05 22:32:34

Trading Partner: Petco

Amount: \$98.10

Dates

Terms

References

Parties

FOBs

Carriers

Adjustments

Do Not Deliver Before2025-02-01

LINE	ITEM INFO	ORDERED QUANTITY	UNIT PRICE	AMOUNT	ACCEPTED QUANTITY	ACCEPTED UNIT PRICE	ACCEPTED AMOUNT	ACCEPT/REJECT REASON
1	Description: GLOVE XLG 11 MIL AMBIDEXTROUS Item Number: 817418020175 Buyer's Item Number: 817418020175 UPC: 817418020175	1	98.10 EA	98.10	1	98.1000	98.10	Reason IA - Accepted Date Type Date mm/dd/yyyy Industry Code

Hide Details

Additional Pricings

Class of Trade Code:RS

Price Identifier Code:RES

Unit Price:24.99

Multiple Price Quantity:1

Manage

elevate

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EDI 850 Document in Elevate

a. Retailer Sends an EDI 850 Purchase Order

This is where everything starts.

A retailer like AutoZone, Walmart, or a regional distributor realizes it needs more products. The retailer's system triggers an EDI 850 PO automatically based on inventory or sales.

What the PO includes:

- Items needed
- Quantities
- Pricing
- Requested delivery dates
- Shipping location or store DC
- Special instructions (labeling, carton packing, routing guide requirements)

Manufacturers do not want to type this into the ERP manually, so EDI loads the order automatically.

b. Manufacturer Confirms What They Can Produce

Once the 850 hits the EDI system, the supplier sends an 855 Purchase Order Acknowledgment.

Three scenarios often happen:

- Yes, we can ship it all
- We can ship part of it now
- We need to backorder certain items

This gives the customer immediate visibility into what is realistic instead of waiting for surprises.

c. Manufacturing Schedules Work Based on the PO

Behind the scenes, that approved PO triggers:

- Bill of materials allocation
- Raw material pull from inventory
- Work orders on the factory floor
- Machine time scheduling
- Labor assignment
- Quality checkpoints

This is the part most people forget:

EDI is not just a file.

It kicks off physical production activity.

d. Goods Move into Finished Inventory

As production completes, the ERP updates finished goods inventory automatically.

No one needs to notify order entry.

No one re-enters data.

No one hopes the spreadsheet matches reality.

EDI keeps the front of the house (sales orders) aligned with the back of the plant (production).

e. Warehouse Picks, Packs, Labels, and Preps Shipment

Once inventory is ready, the warehouse or distribution center receives a pick ticket.

This comes from the ERP, which was triggered by the EDI PO.

For most large retailers, packing and labeling rules matter even more than the carton contents.

Manufacturers must meet requirements such as:

- GS1 labels
- Store-specific pallet breakdowns
- UCC128/GS1-128 labels tied to ASN carton data
- Pack-by-style or pack-by-store instructions
- Serialized or lot-controlled tracking

History of EDI

- **Edward Guilbert** is widely regarded as the father of **Electronic Data Interchange (EDI)**. He introduced the concept of EDI in the 1960s to improve communication and information exchange within supply chains.
- The **U.S. transportation industry** was one of the first sectors to implement EDI, using it to improve communication and coordination among different companies. In **1985**, the **United Nations** introduced **EDIFACT (Electronic Data Interchange for Administration, Commerce, and Transport)**, a global EDI standard that enabled businesses in different countries to exchange electronic business documents using a common format.
- By the mid-1980s, approximately **12,000 companies** in the United States had adopted EDI. The **grocery** and **automobile** industries rapidly embraced EDI because it simplified business transactions, reduced paperwork, minimized errors, and provided a standardized method for exchanging data.
- Today, EDI is widely used by large organizations across the world to exchange business documents quickly, accurately, and securely while complying with established EDI standards.
- **Examples of EDI** include Purchase orders, invoices, shipping statuses, payment information, and so on.

EDI Layered Architecture

EDI layered architecture:-

EDI semantic layer	Application Layer Services	
EDI standard layer	EDIFACT business form standards	
	ANSI X12 business form standards	
EDI transport layer	Electronic mail	X.435, MIME
	Point to point	FTP, TELNET
	World Wide Web	HTTP
Physical layer	Dial-up lines, Internet, I-way	

EDI Layered Architecture

- Semantic (or application) layer
- Standards translation layer
- Packing (or transport) layer
- Physical network infrastructure layer

EDI semantic layer:

- The **EDI Semantic Layer**, also known as the **Application Layer**, describes the business application that generates and processes EDI transactions. It represents the business information in the format used by a company's internal software.
- For a procurement application, this layer includes business documents such as:
 - Request for Quotation (RFQ)
 - Price Quotation
 - Purchase Order (PO)
 - Purchase Order Acknowledgment
 - Invoice
- The semantic layer is **company-specific**, meaning that each organization stores and manages business information according to its own software applications, database structure, and business processes.
- Therefore, the user interface and document formats are customized to the organization's local environment. Before these documents can be exchanged with trading partners, they must be translated into a standard EDI format at the Standards Translation Layer.

EDI Layered Architecture

- **EDI Standard layer(Standards translation):**
 - Specifies business form structure so that information can be exchanged
 - The information seen at the EDI semantic layer must be translated from a company-specific form to a more generic or universal form so that it can be sent to various trading partners, who could be using a variety of software applications at their end.
 - To achieve this, organizations adopt universal EDI standards that specify the acceptable fields, layout, and structure of business documents such as **purchase orders, invoices, and shipping notices**. These standards ensure that business documents are interpreted consistently by all trading partners.
 - The two major EDI standards are:
 - **ANSI X12 (American National Standards Institute X12):**
 - Developed by the **American National Standards Institute (ANSI)**.
 - Primarily used in **North America**, especially in the **United States and Canada**.
 - **EDIFACT (Electronic Data Interchange for Administration, Commerce and Transport):**
 - Developed by the **United Nations Economic Commission for Europe (UN/ECE)**.
 - Used internationally to support global electronic business transactions.
 - The existence of these two major standards can make EDI implementation more complex because different trading partners may use different standards. Therefore, **EDI translation software** is used to convert documents from one standard to another, ensuring seamless and accurate communication between businesses.

EDI Layered Architecture

- **EDI transport layer**
- The **EDI Transport Layer**, also known as the **Packing (or Transport) Layer**, is responsible for transmitting business documents from one trading partner to another.
- It determines **how** the business documents are sent across the communication network.
- Traditionally, business documents were delivered through methods such as:
 - Postal mail
 - Courier services (e.g., UPS)
 - Fax
- Today, **e-mail** and other electronic communication networks are the most commonly used carriers for EDI messages because they provide faster, more reliable, and cost-effective communication.
- Although EDI documents may be transmitted through e-mail, **EDI is different from ordinary e-mail**. A regular e-mail is intended for human reading, whereas EDI is designed for **computer-to-computer communication** without manual intervention.

EDI Layered Architecture

- EDI document transport is far more complex than simply sending e-mail messages or sharing files through a network.
- These EDI documents are more structured than e-mail.
- In addition, EDI messages have certain legal status.
- When a trading partner sends a business document, **EDI translation software** converts the company's proprietary (internal) document format into a **standard EDI format** agreed upon by both trading partners. The standardized document is then transmitted through the transport layer.
- After the receiving company gets the document, its **EDI translation software** automatically converts the standard EDI format back into the company's proprietary internal format. This allows the organization's business applications to process the document automatically without re-entering the data manually.
- For instance, if a buyer sends a supplier EDI purchase orders that specify the requirements, time of delivery, and quantity and the supplier does not uphold its end of the contract, it can be taken to court with the EDI trading agreements serving as evidence.
- Table below indicates some EDI properties which distinguish it from e-mail

Electronic Data Interchange versus E-mails

Electronic Data Interchange (EDI)

There is typically no human involvement in the processing of the information, as the interface has software-to-software orientation. The data are structured in a software-understandable way.

The interchange is composed by one software for interpretation by another software. If a reply is involved, it is composed by a software to be interpreted by another software.

Electronic Mail

The data are not necessarily structured to be software-understandable. A human-to-software interface is involved at a minimum of one end of the interchange.

The message is composed by a human and/or interpreted by a human and/or a reply is composed by a human and/or interpreted by a human.

EDI Applications in various fields of Business

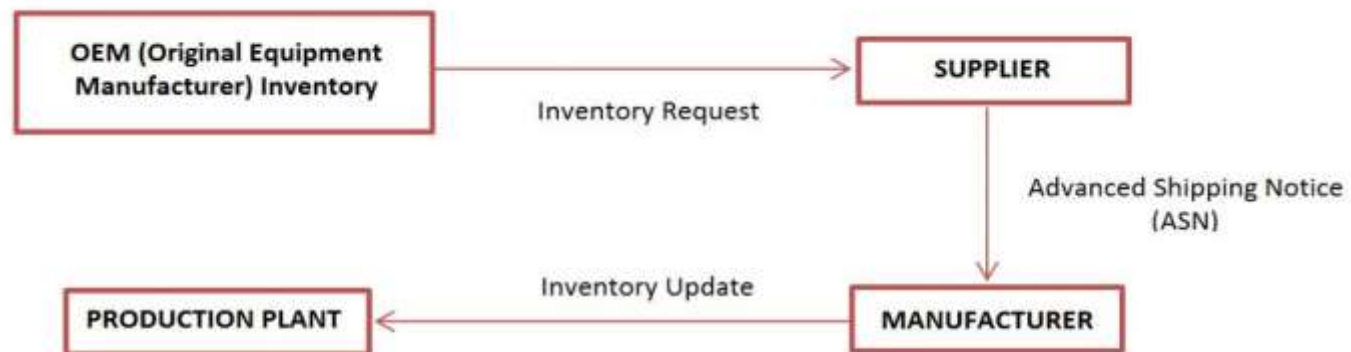
- **Retail Sector** – In the retail sector profit margins usually depend upon efficient inventory management. EDI provides a structured way to maintain and replenish goods stocked at a retail outlet. Retailers use a common model stock for each shop location and the point of sale stock position is updated continuously and data is fed via EDI enabled SCM (supply chain management) network. The EDI software monitors all the logistics and makes updates in the original stock.



EDI in RETAIL INDUSTRY

EDI Applications in various fields of Business

- **Manufacturing Sector** – EDI ensures effective and efficient management of materials required for production of a commodity. In manufacturing sector EDI facilitates Material requirement planning and just in time manufacturing. The Inventory position of OEM is constantly updated through EDI and the supplier is notified about shortage of materials. This helps the supplier to plan and schedule supply according to requirements of the manufacturer. The suppliers respond via EDI with an ASN to identify the parts/materials to be delivered and the approximate delivery time and as soon as the shipment is delivered at the production plant the inventory is updated again.



EDI in MANUFACTURING INDUSTRY

EDI Applications in various fields of Business

- **Automobile Sector** – In automobile sector EDI is used to keep customers updated with the current product and pricing information during the purchase cycle. An advance shipping notice is transmitted through EDI to the customers to prepare a loading schedule and to ensure proper receipt of the product. The customer may also make payment on receipt of goods via EDI to speed up the payment process.
- **Order Processing:** EDI streamlines the exchange of purchase orders and order confirmations between buyers and suppliers. This automation speeds up the ordering process, reduces errors, and ensures that orders are processed efficiently.
- **Invoicing and Billing:** EDI is used to generate and send invoices electronically. This accelerates the billing cycle, improves accuracy, and facilitates faster payment processing.
- **Shipping and Logistics:** EDI is employed in the exchange of shipping instructions, bills of lading, and shipment status updates. This enhances supply chain visibility and coordination, reducing transit times and minimizing errors.
- **Retail and E-commerce:** Many retail companies and e-commerce platforms use EDI to automate the exchange of product information, pricing, and sales orders with suppliers. It ensures that product listings and inventory levels are up to date.

EDI Applications in various fields of Business

- **Manufacturing and Supply Chain:** EDI supports the efficient flow of information between manufacturers, suppliers, and distributors, helping to synchronize production schedules, manage demand, and reduce lead times.
- **Food and Beverage Industry:** EDI assists with order management, inventory tracking, and compliance with food safety regulations. It helps food manufacturers and distributors maintain traceability and quality control.
- **Energy and Utilities:** Utility companies use EDI for meter reading data, billing, and energy consumption reporting. This supports efficient customer billing and resource management.

Security and Privacy issues of EDI

- Since in the case of EDI, we are dealing with trade between countries and corporations, issues of legal admissibility and computer security are important.
- Companies that deal with EDI often retain the services of a lawyer during the design of an EDI application so that the appropriate evidentiary/admissibility safeguards are implemented.
 - Legal Status of EDI Messages
 - Digital Signatures and ED

- **Legal Status of EDI Messages :**

- There has been considerable debate concerning the legal status of EDI messages and electronic messages in general.
- Although a lot of work is being done on legal framework, nothing concrete has come out these efforts.
- No rules exist that indicate how electronic messages may be considered binding in business or other related transactions
- The establishment of such a framework is essential if EDI is to become widespread

- **Digital Signatures and EDI :**

- The cryptographic community is exploring various technical uses of digital signatures by which messages might be time-stamped or digitally notarized to establish dates and times at which a recipient might claim to have had access or even read a particular message
- If digital signatures are to replace handwritten signatures, they must have the same legal status as handwritten signatures (documents signed with digital signatures must be legally binding).
- Digital signatures should have greater legal authority than handwritten signatures.
- For instance, if a ten-page contract is signed by hand on the tenth page, one cannot be sure that the first nine pages have not been altered.
- If the contract was signed by digital signatures, however, a third party can verify that not one byte of the contract has been altered.

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E-commerce and Industry Value Chain

Industry Structure

- **Industry structure** describes the overall business environment in which firms operate and the general profitability of an industry. It refers to the nature of the competitors, suppliers, buyers, and other market forces that influence competition and business success. Rivalry among existing competitors,
- According to **Porter's Five Forces Model**, an industry's structure is determined by the following five competitive forces:
- **Rivalry Among Existing Competitors:**
 - The level of competition among firms already operating in the industry.
 - High rivalry leads to price competition, increased marketing costs, and lower profits.
- **Threat of Substitute Products:**
 - The possibility that customers will switch to alternative products or services that satisfy the same need.
 - A high threat of substitutes limits the prices that companies can charge.

E-commerce and Industry Value Chain

- **Barriers to Entry:**
 - Factors that make it difficult for new firms to enter the industry, such as high investment costs, government regulations, strong brand loyalty, or advanced technology.
 - High barriers protect existing firms from new competitors. If new companies can enter easily, competition increases.
- **Bargaining Power of Suppliers:**
 - The ability of suppliers to influence the price and quality of raw materials or services.
 - When suppliers have strong bargaining power, they can increase prices or reduce product quality, lowering industry profitability.
- **Bargaining Power of Buyers:**
 - The ability of customers to negotiate lower prices, demand better quality, or switch to competitors.
 - Strong buyer power reduces the profitability of firms in the industry.
- E-commerce significantly changes industry structure by altering the relative strength of Porter's five competitive forces. It enables businesses to reach global markets, lowers entry barriers for new firms, increases competition, provides buyers with more choices and information, and changes relationships between suppliers and buyers. As a result, e-commerce reshapes the competitive environment and affects the profitability of industries.

Porter's Five Forces





Figure: How E-Commerce Influences Industry Structure

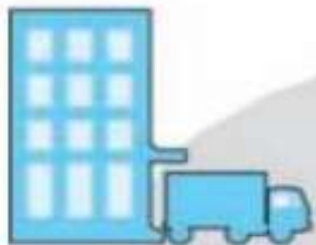
- When you consider a business model and its potential long-term profitability, you should always perform an industry structural analysis.
- An industry structural analysis is an effort to understand and describe the nature of competition in an industry, the nature of substitute products, the barriers to entry, and the relative strength of consumers and suppliers.
- E-commerce can affect the structure and dynamics of industries in very different ways.
- In the travel industry, entirely new middlemen such as Travelocity entered the market to compete with traditional travel agents.
- After Travelocity, Expedia, CheapTickets, and other travel services demonstrated the power of e-commerce marketing for airline tickets, the actual owners of the airline seats—the major airlines—banded together to form their own Internet outlet for tickets, Orbitz, for direct sales to consumers (although ultimately selling the company to a private investor group).
- Clearly, e-commerce creates new industry dynamics that can best be described as the give and take of the marketplace, the changing fortunes of competitors.
- Similarly, In the chemical and automobile industries, e-commerce is being used effectively by manufacturers to strengthen their traditional distributors. In these industries, e-commerce technology has not fundamentally altered the competitive forces—bargaining power of suppliers, barriers to entry, bargaining power of buyers, threat of substitutes, or rivalry among competitors—within the industry.
- Hence, each industry is different and you need to examine each one carefully to understand the impacts of e-commerce on competition and strategy.

- Inter-firm rivalry (competition) is one area of the business environment where ecommerce technologies have had an impact on most industries.
- In general, e-commerce has increased price competition in nearly all markets. It has been relatively easy for existing firms to adopt e-commerce technology and attempt to use it to achieve competitive advantage vis-à-vis rivals.
- For instance, e-commerce inherently changes the scope of competition from local and regional to national and global. Because consumers have access to global price information, ecommerce produces pressures on firms to compete by lowering prices (and lowering profits).
- On the other hand, e-commerce has made it possible for some firms to differentiate their products or services from others. Amazon patented one-click purchasing, for instance, while eBay created a unique, easy-to-use interface and a differentiating brand name.
- Therefore, although e-commerce has increased emphasis on price competition, it has also enabled businesses to create new strategies for differentiation and branding so that they can retain higher prices.

- **INDUSTRY VALUE CHAINS**

- One of the basic tools for understanding the impact of information technology on industry and firm operations is the industry value chain.
- While an industry structural analysis helps you understand the impact of e-commerce technology on the overall business environment in an industry, a more detailed industry value chain analysis can help identify more precisely just how e-commerce may change business operations at the industry level.
- A value chain is the set of activities performed in an industry or in a firm that transforms raw inputs into final products and services. Each of these activities adds economic value to the final product; hence, the term value chain as an interconnected set of value-adding activities.
- Figure below illustrates the six generic players in an industry value chain:
 - 1. Suppliers,
 - 2. Manufacturers,
 - 3. Transporters,
 - 4. Distributors,
 - 5. Retailers, And
 - 6. Customers.

SUPPLIERS



MANUFACTURERS



Supply Chain
Management
Systems

DISTRIBUTORS



Inventory
Management
Systems

RETAILERS



Efficient
Customer Response
Systems

CUSTOMERS



TRANSPORTERS

Transportation Management Systems

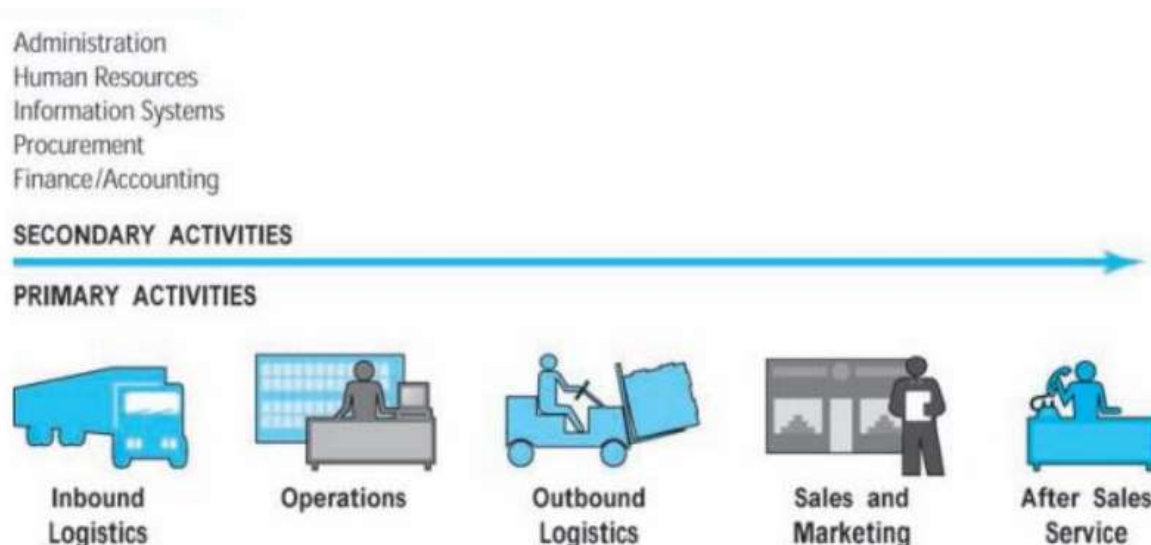
Alternative Direct Channels
e.g., the Web



- By reducing the cost of information, e-commerce offers each of the key players in an industry value chain new opportunities to maximize their positions by lowering costs and/or raising prices. For instance, manufacturers can reduce the costs they pay for goods by developing Internet-based B2B exchanges with their suppliers. Manufacturers can develop direct relationships with their customers, bypassing the costs of distributors and retailers.
- Distributors can develop highly efficient inventory management systems to reduce their costs, and retailers can develop highly efficient customer relationship management systems to strengthen their service to customers. Customers in turn can search for the best quality, fastest delivery, and lowest prices, thereby lowering their transaction costs and reducing prices they pay for final goods. Finally, the operational efficiency of the entire industry can increase, lowering prices and adding value for consumers, and helping the industry to compete with alternative industries.

FIRM VALUE CHAINS

- The concept of value chain can be used to analyse a single firm's operational efficiency as well. The question here is: How does e-commerce technology potentially affect the value chains of firms within an industry?
- A firm value chain is the set of activities a firm engages in to create final products from raw inputs.
- Each step in the process of production adds value to the final product. In addition, firms develop support activities that coordinate the production process and contribute to overall operational efficiency.
- Figure below illustrates the key steps and support activities in a firm's value chain.



- E commerce offers firms many opportunities to increase their operational efficiency and differentiate their products. For instance, firms can use the Internet's communications efficiency to outsource some primary and secondary activities to specialized, more efficient providers without such outsourcing being visible to the consumer.
- In addition, firms can use ecommerce to more precisely coordinate the steps in the value chains and reduce their costs.
- Finally, firms can use e-commerce to provide users with more differentiated and high-value products.
- For instance, Amazon provides consumers with a much larger inventory of books to choose from, at a lower cost, than traditional book stores.
- It also provides many services—such as instantly available professional and consumer reviews, and information on buying patterns of other consumers—that traditional bookstore cannot.

FIRM VALUE WEBS

- While firms produce value through their value chains, they also rely on the value chains of their partners—their suppliers, distributors, and delivery firms. E-commerce creates new opportunities for firms to cooperate and create a value web.
- A value web is a networked business ecosystem that uses e-commerce technology to coordinate the value chains of business partners within an industry, or at the first level, to coordinate the value chains of a group of firms.
- A value web coordinates a firm's suppliers with its own production needs using an Internet-based supply chain management system. Firms also use the Internet to develop close relationships with their logistics partners. For instance, Amazon relies on UPS tracking systems to provide its customers with online package tracking, and it relies on the U.S. Postal Service systems to insert packages directly into the mail stream. Amazon has partnership relations with hundreds of firms to generate customers and to manage relationships with customers. In fact, when you examine Amazon closely, you realize that the value it delivers to customers is in large part the result of coordination with other firms and not simply the result of activities internal to Amazon. The value of Amazon is, in large part, the value delivered by its value web partners. This is difficult for other firms to imitate in the short run.
- Figure below illustrates a value web.

STRATEGIC ALLIANCE AND PARTNER FIRMS

